

SOCIOL 723: Social Statistics II

Week 6, Matching, Propensity Scores, and Weighting

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Today

Matching

The Propensity Score

Weighting

Doubly Robust Estimation

The Anatomy of Adjustment

Selection on Unobservables: Heckman

★ = essential, you must understand this

★ = for understanding, not examined

Where We Are

Last week we established that under **conditional ignorability**, **positivity** (today also called *overlap* or *common support*), **consistency**, and **SUTVA**,

$$\text{ATE} = E\left[E[Y_i \mid D_i = 1, X_i] - E[Y_i \mid D_i = 0, X_i]\right].$$

Today we take that identification result as *given* and ask a purely practical question: **how should we actually do the adjustment?**

Four answers, and they are more alike than they look

1. **Matching:** find comparison units that look like the treated ones.
2. **Regression / g-computation:** model $E[Y \mid D, X]$ and average.
3. **Weighting:** reweight the sample so treatment is unrelated to X .
4. **Doubly robust:** do both, and get insurance.

All four estimate the *same* identified quantity. They differ in what they model, **what weights they put on covariate cells**, what they extrapolate, and how they fail.

What Regressing Y on D (and X) Gives ★

On D alone. One dummy is a saturated model, so OLS returns the CEF $E[Y_i | D_i]$ at 1 and 0: the coefficient on D_i is the naive comparison $\bar{Y}_1 - \bar{Y}_0$. With $\tau = \text{ATE}$ and $\pi = \Pr(D_i = 1)$,

$$\begin{aligned} E[Y_i | D_i = 1] - E[Y_i | D_i = 0] - \tau \\ = \underbrace{E[Y_i(0) | D_i = 1] - E[Y_i(0) | D_i = 0]}_{\text{baseline selection}} + \underbrace{(1 - \pi)(\text{ATT} - \text{ATU})}_{\text{selection on the effect}} \end{aligned}$$

On D and X , under conditional ignorability. Within a cell $X_i = x$, treatment is as good as random: the treated and the untreated share the same distribution of baselines *and* of gains. Both terms vanish cell by cell, and

$$E[Y_i | D_i = 1, X_i = x] - E[Y_i | D_i = 0, X_i = x] = \tau(x) \quad \text{Week 5's CATE}(x)$$

Selection is gone. Heterogeneity is not gone; it is now a set of numbers $\tau(x)$, one per cell, waiting to be averaged.

Three Targets, One Set of Cell Effects ★

The $\tau(x)$ are fixed by the data and ignorability. An estimand is a rule for averaging them; each rule answers a different question:

$$\text{ATE} = \sum_x \tau(x) \Pr(X_i = x) \quad \text{everyone: a universal policy}$$

$$\text{ATT} = \sum_x \tau(x) \Pr(X_i = x \mid D_i = 1) \quad \text{the treated: what treatment did for them}$$

$$\text{ATU} = \sum_x \tau(x) \Pr(X_i = x \mid D_i = 0) \quad \text{the untreated: what it would do for them}$$

with $\text{ATE} = \pi \text{ATT} + (1 - \pi) \text{ATU}$ (Week 5).

- College: did graduates gain (ATT)? Would non-graduates (ATU)? Would everyone (ATE)? Under selection on gains they differ.
- The weights are covariate distributions: the ATT counts a cell by how common it is *among graduates*.
- Name the target first (Week 5). Every method today computes one of these, or, unnoticed, a different one.

What OLS Returns ★

OLS of Y_i on D_i and a saturated X_i returns *one* coefficient. It too is an average of the $\tau(x)$, but with weights OLS picks, where $e(x) = \Pr(D_i = 1 \mid X_i = x)$:

$$\tau_R = \frac{\sum_x \tau(x) e(x)(1 - e(x)) \Pr(X_i = x)}{\sum_x e(x)(1 - e(x)) \Pr(X_i = x)} \quad \text{weight: the variance of treatment in the cell}$$

- τ_R is none of the three targets unless $\tau(x)$ is constant. It over-weights coin-flip cells and ignores lopsided ones: **a number without a named target**. GSS, college on log earnings: naive 0.683, $\tau_R = 0.631$, ATE 0.635, ATT 0.625.
- **Ignorability settles identification; it does not choose the weights**. Where OLS's weights come from is derived at the end of the lecture.

Matching: take the weights back

Compute $\tau(x)$ cell by cell and average with the weights of the target you named: the next section. The rest of the lecture is what to do when cells run out.

Matching

Exact Matching for the ATE ★

Slide 5's ATE with sample quantities. Cells are the values of X_i ; in every cell that holds both treated and untreated units,

$$\hat{\tau}(x) = \bar{Y}_{1x} - \bar{Y}_{0x}, \quad \widehat{ATE} = \sum_x \frac{n_x}{n} \hat{\tau}(x) \quad n_x \text{ units in cell } x$$

- Each treated unit is compared only with untreated units that share its X_i : “like with like”, nothing modelled. Under ignorability $\hat{\tau}(x)$ estimates $\tau(x)$, and the weights are the population's.
- The requirement is visible: **a cell needs both arms**. Where it lacks one, $\hat{\tau}(x)$ does not exist. Next frame.

Unmatched Cells ★

A cell with treated units and no untreated ones, or the reverse, has no $\hat{\tau}(x)$. Three responses, in order of honesty:

1. **Drop the cell and say so.** The estimand becomes the ATE on the matched population, $\sum_x \tau(x) \Pr(X_i = x \mid \text{matched})$. Report the share of units, and of treated units, that were dropped, and what those cells look like.
2. **Coarsen.** Merge the cell with its neighbours (five-year age bands, education groups). Merged cells hold both arms more often, at the price of comparing units that differ within a band: a bias you then check (two frames on).
3. **Model.** Predict $\bar{Y}_{0x}$ for the cell from nearby cells. This is regression's answer, and it is extrapolation: right only if the model is, and untestable where the data are absent.

Positivity fails in the sample before it fails in the population. An empty cell is not an error message; it is the estimand quietly changing. Matching's merit is that it makes you look.

What You Check, for the ATE ★

- **Coverage.** How many treated and untreated units sit in matched cells, and which cells were lost. This is the common-support check in its most transparent form.
- **Balance inside the cells.** Variables matched on exactly agree within every cell by construction. Variables you coarsened do not: inside a cell, graduates may still have more parental education than non-graduates. For each such variable take the treated–untreated gap within each matched cell, average the gaps over cells with the ATE weights n_x/n , and divide by the pooled SD: the *standardized mean difference*, with 0.1 the usual bar. Do the same for variables you did not match on at all.
- **Representativeness.** Compare the matched population's cell shares with the full sample's. If dropping cells changed whom you describe, say so.

Matching for the ATT ★

Now the treated are the target population, and the operation changes:

$$\widehat{\text{ATT}} = \sum_x \frac{n_{1x}}{n_1} \hat{\tau}(x) \quad n_{1x} \text{ treated in cell } x, n_1 \text{ treated in matched cells}$$

- Keep every treated unit. In each treated cell, weight the untreated units so that, cell by cell, the untreated look like the treated. The treated themselves are never reweighted.
- **Asymmetry.** Untreated units in cells with no treated unit go unused at no cost. Treated units in cells with no untreated unit are the problem: dropping them makes it the effect on the *matchable* treated.
- **The checks follow.** Coverage: share of treated dropped. Balance: reweighted untreated against the treated, the treated SD as yardstick. Variance: how many untreated units each treated cell draws on.
- Positivity is needed only where the treated are: $e(x) < 1$ suffices.

A Real Application: College and Earnings in the GSS ★

A bachelor's degree (1,441 of 3,509 full-time workers 25–64, GSS 2010–22) on log earnings. Five covariates coarsened into cells:

female, black	as they are	possible cells	108
parental education	<12, 12–13, 14+ years	occupied	106
vocabulary score	0–5, 6–7, 8–10 correct	with both arms	92
experience	0–16, 17–29, 30+ years	treated only / untreated only	2 / 12
		treated dropped	4 of 1,441
		untreated dropped	70 of 2,068

Coverage is nearly complete: 74 units and 4 treated lost, so the matched estimands are essentially the full-sample ones. Ten of the twelve untreated-only cells are Black respondents whose parents did not finish high school: where graduates are scarcest, the data cannot compare. (figs/w6_matching_gss.R)

A Real Application: the Estimates ★

	estimate
naive $\bar{Y}_1 - \bar{Y}_0$	0.683
ATE, matched cells	0.635
ATT, matched cells	0.625
ATU, matched cells	0.642
OLS on D and 106 cell dummies, τ_R	0.631

- The same 92 numbers $\hat{\tau}(x)$ under four sets of weights. Adjustment removes 0.05 of the 0.68 raw gap: within these cells, selection into college on parental education and scores explains only part of the graduate premium.
- ATT below ATU: graduates gain a little less than non-graduates would, the direction Brand and Xie (2010) report, though the gap is small.
- τ_R sits between, as slide 6 said. Effects this homogeneous make the weights matter little; the point is that you can tell.

A Real Application: the Checks ★

Within-cell treated–untreated gaps, averaged over matched cells with the ATE weights (n_x/n) or the ATT weights (n_{1x}/n_1), divided by the pooled SD:

	raw	ATE-matched	ATT-matched
parental education (coarsened)	0.736	0.125	0.141
vocabulary score (coarsened)	0.793	0.123	0.109
experience (coarsened)	−0.420	−0.108	−0.093
female (exact)	0.111	0.000	0.000
black (exact)	−0.150	0.000	0.000
other race (not matched on)	−0.052	0.070	0.095

- Exact variables: zero by construction. Coarsened ones: from 0.7 to about 0.1, the within-band imbalance coarsening left behind; finer bands might remove it.
- Other race got *worse*: matching balances what you match on and nothing else.

The Curse of Dimensionality ★

The GSS example needed 108 cells for five coarsened covariates and still lost 14. With p binary covariates there are 2^p cells: $p = 10$ gives 1,024, $p = 20$ over a million. Exact matching becomes impossible long before the covariates run out.

Practical responses, in rough order of desperation:

- **Coarsened exact matching** (CEM): bin continuous covariates coarsely, age in five-year bands, income in quintiles, then match exactly on the bins.
Transparent, and you see immediately what you had to discard.
- **Nearest-neighbour matching** on a distance: Euclidean, or *Mahalanobis* $[(X_i - X_j)' \hat{\Sigma}^{-1} (X_i - X_j)]^{1/2}$, which scales by the covariance so no variable dominates by its units.
- **Propensity score matching**: collapse X to a single number, the next section.

The Curse of Dimensionality ★ (cont.)

Every one of these is a way of admitting that exact comparisons run out in *your sample*. That is finite-sample sparsity, which is not the same thing as a structural positivity failure ($e(x) \in \{0, 1\}$ in the population), though in practice you often cannot tell them apart. Being explicit about where the comparisons thinned is a feature.

Matching Is Preprocessing, Not Estimation ★

Ho, Imai, King, and Stuart (2007) make the point that should organize your workflow: matching is a *nonparametric preprocessing step* that reduces how much your final model has to do.

1. Match, using **no outcome data at all**. This is the design stage; it can be iterated freely without *p*-hacking, because you cannot see the answer.
2. Check **coverage and balance**, as above: standardized mean differences below 0.1, variance ratios near 1, density overlaps.
3. *Then* run your estimate on the matched sample.

In the Literature: Mazrekaj, De Witte, and Cabus (2020)

The question. Do children raised by same-sex parents do worse in school? Census-based studies found a negative association or none, and could not tell children *born* to same-sex parents from children who happened to live with them at one moment.

The data. Dutch registers: every child born 1998–2007 who reached the last year of primary school by 2019, $n = 1,204,692$. Same-sex parents are read off household composition. Three treated groups:

children who ever lived with same-sex parents	2,971 (0.25%)
born to same-sex parents, never adopted or fostered	1,390
adopted or fostered	50
from a parent's previous different-sex relationship	1,531

The outcome. The standardized test at the end of primary school, age 12, scaled within test and year to mean 0, SD 1.

Mazrekaj et al.: the Selection You Can See

Their Table 1, covariates measured *at the child's birth*, not at the test (measured later they would be post-treatment, Week 5):

	different-sex	same-sex, any	same-sex, from birth
household net income (EUR)	79,300	86,128	90,779
at least one parent upper-secondary	.897	.931	.960
parents' average age	32.7	34.1	36.1
number of children in household	1.88	1.60	1.53
parents married	.772	.536	.649
children	1,201,721	2,971	1,390

- Selection runs toward advantage: richer, older, better-educated parents, smaller families. Having a child as a same-sex couple is costly and deliberate, so a raw comparison overstates any benefit.
- The treated are one in four hundred: almost every untreated cell will go unused, and what matters is whether every treated cell finds a match.

Mazrekaj et al.: OLS, Then CEM

Their Table 2, outcome the end-of-primary test score, standard errors clustered by neighbourhood:

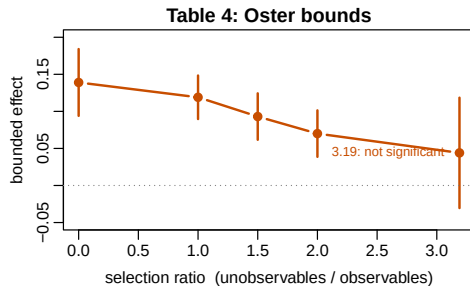
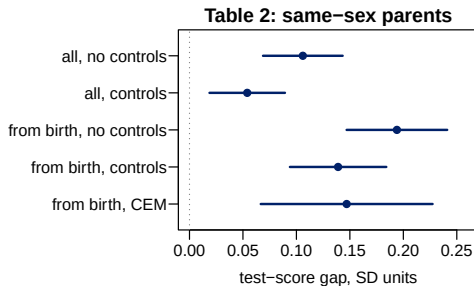
	ever with same-sex parents		born to same-sex parents		
	(1)	(2)	(3)	(4)	(5) CEM
same-sex parents	.106 (.019)	.054 (.018)	.194 (.024)	.139 (.023)	.147 (.041)
controls at birth, nbhd FE	no	yes	no	yes	matched
children	1,204,692	1,204,692	1,195,624	1,195,624	48,388
treated	2,971	2,971	1,390	1,390	1,390

- Controls halve the coefficient: about half the raw gap was Table 1's selection. Restricting to children born to same-sex parents raises it: the mixed group carried the scar of parental separation.
- CEM and OLS on the same covariates agree, .147 against .139; the price of pruning a million controls is the wider interval.

Mazrekaj et al.: What CEM Did

- **Coarsen.** Income to deciles; parents' average age to six bands; family size to 1, 2, 3+. Sex, birth year, ethnicity, parental education, family structure, and neighbourhood were already categorical.
- **Match exactly on every covariate.** A cell is one combination of all of them. Each of the 1,390 treated children is compared only with untreated children in its own cell.
- **Prune.** Every untreated child who shares a cell with a treated child is kept, with no limit per treated child: 46,998 of them, about 34 per treated child. The million set aside sit in cells with no treated child, because the cells include neighbourhood at birth and birth year. Our “untreated-only cells go unused”, at scale. Every treated child found a cell, so the estimand is the ATT on all 1,390.
- **Why also OLS.** Agreement is a check of *model dependence*: the linear specification was not doing the work. It says nothing about unobservables; that is the bounds, next.

Mazrekaj et al.: Bounds and Secondary School



- **Bounds** (their Table 4, Oster 2019). Unobserved selection would have to be 3.19 times the observed selection to make the effect insignificant; at twice, the gap is still .070. The sensitivity frame's question, in regression form.
- **Secondary school** (their Table 5, cohorts 1998–2001, 280 treated): +4.8 points of diploma attainment on a base of 92% with controls, +3.8 with CEM.

Mazrekaj et al.: Lessons for Your Own Matching

1. **Define the treatment before you match.** “Ever lived with” and “born to” are different populations with different answers; the first mixes in the effect of separation.
2. **Measure covariates before the treatment,** here at birth. Income or neighbourhood at the test would be post-treatment (Week 5).
3. **Show the selection,** as their Table 1 does, so the reader knows which way the raw gap is biased.
4. **Report what pruning did:** cells kept, controls set aside, whether any treated unit was lost.
5. **Regression beside matching** checks model dependence, not ignorability. The causal claim rests on the bounds, and on saying what they assume: that observed selection is informative about unobserved selection.

Propensity Score

A Remarkable Theorem ★

Define the **propensity score**

$$e(x) \equiv \Pr(D_i = 1 \mid X_i = x).$$

Rosenbaum and Rubin (1983)

If $(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i \mid X_i$, then

$$(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i \mid e(X_i).$$

- Conditioning on a *single number* suffices. A p -dimensional matching problem collapses to a one-dimensional one.
- $e(X)$ is a *balancing score*: within strata of $e(X)$, the whole covariate vector is distributed identically in the two groups.
- This is why the propensity score is everywhere. It is also why it is routinely oversold; it does nothing about *unobserved* confounders.

Proof of the Balancing Property ★

It suffices to show $\Pr(D_i = 1 \mid X_i, e(X_i)) = \Pr(D_i = 1 \mid e(X_i))$; the left side is $e(X_i)$ by definition. For the right side, use iterated expectations:

$$\Pr(D_i = 1 \mid e(X_i)) = E[E[D_i \mid X_i] \mid e(X_i)] = E[e(X_i) \mid e(X_i)] = e(X_i).$$

So $D_i \perp\!\!\!\perp X_i \mid e(X_i)$. Now the potential outcomes, by the same move:

$$\begin{aligned} \Pr(D_i = 1 \mid Y_i(0), Y_i(1), e(X_i)) \\ &= E[\Pr(D_i = 1 \mid Y_i(0), Y_i(1), X_i) \mid Y_i(0), Y_i(1), e(X_i)] && \text{iterated expectations} \\ &= E[e(X_i) \mid Y_i(0), Y_i(1), e(X_i)] = e(X_i) && \text{ignorability given } X_i \end{aligned}$$

which is $\Pr(D_i = 1 \mid e(X_i))$: given the score, the potential outcomes say nothing more about treatment. ■

- Note this is a statement about the *true* $e(x)$. In practice we estimate it, and the properties of $\hat{e}$ matter, Week 13.
- Any function of X that is finer than $e(X)$ is also a balancing score; $e(X)$ is the coarsest one.

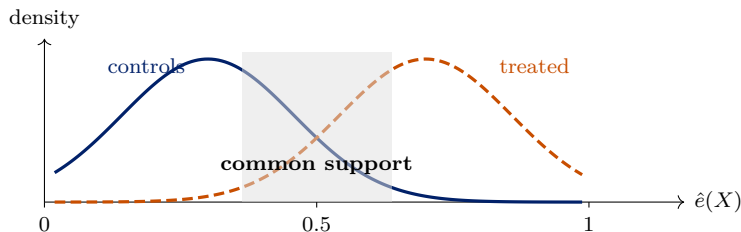
Estimating and Checking It ★

- Usually a logit of D on X (Week 4). **The model is not judged by fit.** A propensity model with a high AUC is often a bad sign, it means treated and control units are easy to tell apart, i.e. poor overlap.
- Judge it by **balance**: does weighting or matching on $\hat{e}$ equalize the covariate distributions? Iterate on the specification until it does, adding interactions and polynomials as needed.
- Plot the two distributions of $\hat{e}$. Mass near 0 among the treated, or near 1 among the controls, is a positivity failure staring at you.

The propensity score paradox

The better your propensity model predicts treatment, the less common support you have, and the less the data can tell you. Prediction quality and identification pull in opposite directions.

Overlap: What to Do When It Fails ★



- **Trim** to a range of $\hat{e}$ (say $[0.1, 0.9]$), and say so. You have changed the estimand to that subpopulation.
- **Redefine the population** to the region where comparison is possible, often the honest and more interesting move.
- **Change the estimand:** the ATT needs overlap only for the treated; the “overlap-weighted” estimand (Li, Morgan, and Zaslavsky 2018), defined in the next section, deliberately targets the region where comparison is best.

In the Literature: Cheng et al. (2021, *Science Advances*)

Cheng, Brand, Zhou, Xie, and Hout. **The question.** Who gains most from a college degree, and when? The *positive selection* hypothesis says those most likely to complete college gain most; the *negative selection* hypothesis (Brand and Xie 2010, Week 5) says the least likely gain most. And a point-in-time premium may miss returns that build over a career.

The data. National Longitudinal Survey of Youth 1979, waves 1979–2014. Respondents aged 14–17 in 1979 ($n = 5,582$), so that every predictor of college is measured before college; at least 12th grade completed ($n = 4,548$). Treatment: a bachelor's degree by age 25, about one in four. Outcome: inflation-adjusted log hourly wage at every age from 25 to 55; person-years with no wage are not set to zero but predicted from the person's own trajectory.

The plan: estimate the propensity of completing college, build a matched sample for the ATE, fit growth curves on it, and let the college effect vary with the score. In that order, then the checks and the results.

Cheng et al.: Building the Propensity Score

The score is a logit, but its specification was *searched*, blind to wages, with balance as the goal (our “the model is not judged by fit”):

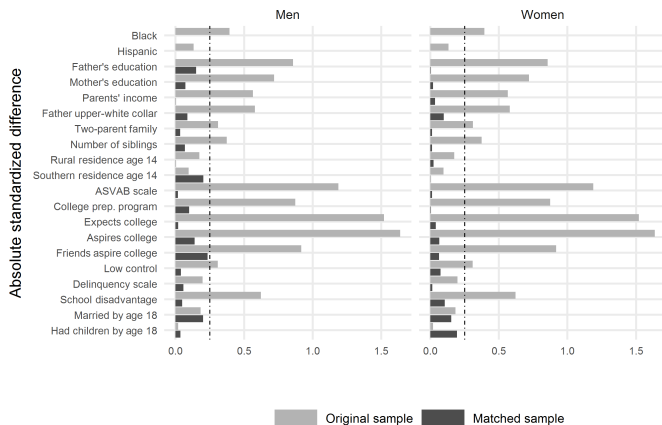
1. **A baseline set that always stays in:** sex, race, parents' education, family income in 1979, both biological parents present, siblings, rural and Southern residence, ability (1980 ASVAB).
2. **Candidates added one at a time** when the likelihood-ratio statistic exceeds 1.0: college-prep curriculum, aspirations and expectations, locus of control, self-esteem, delinquency, school composition, marriage and childbearing by 18. 176 logits; 22 covariates survive; self-esteem and religion do not.
3. **Squares and interactions**, same rule at 2.71: 3,527 logits; one square (mother's education) and 11 interactions survive, e.g. ability $\times$ college-prep programme.

Then trim to common support: 443 non-graduates with $\hat{e} \leq 0.004$ and 20 graduates with $\hat{e} \geq 0.923$ removed.

Cheng et al.: the Matching

- **Exact on race and ethnicity, then nearest neighbour on the score,** separately for men and women. The distance is on the *linear* score $\log[\hat{e}/(1 - \hat{e})]$: on the raw scale a treated unit at 0.10 looks as close to a control at 0.15 as to one at 0.05, but in covariates it is much closer to the first.
- **One to five, with replacement.** Each graduate gets the five closest non-graduates of the same race, each carrying weight $w_i/5$, w_i the graduate's survey weight; the same control can serve many graduates.
- **In both directions.** Each *non-graduate* likewise gets five graduates. Matching graduates to non-graduates alone would give the ATT; matching both ways reweights both arms to the whole population, so the target is the **ATE**, exactly the difference between our ATT and ATE frames.
- **The matched sample.** Units appearing several times are collapsed by summing weights: 2,079 men (423 graduates, 1,656 non-graduates) and 2,045 women (432 and 1,613).

Cheng et al.: the Balance Check (their Figure 1)

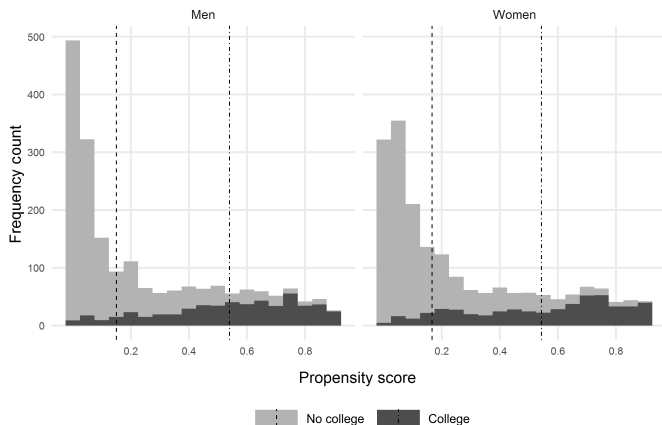


For every covariate in the score: the absolute standardized mean difference between graduates and non-graduates, divided by the SD among graduates, before (light) and after (dark) matching.

The dashed line is 0.25, their bar. Raw gaps of 1–1.7 SD on ASVAB, aspirations, and expectations fall below it.

(Cheng et al. 2021, Fig. 1, CC BY-NC)

Cheng et al.: the Overlap Check (their Figure 2)



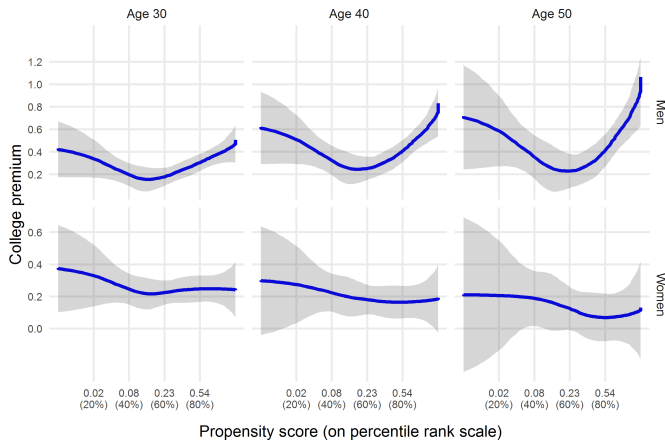
Stacked histograms of $\hat{e}$ by completion; dashed lines at the group means, 0.15 vs 0.54 for men and 0.17 vs 0.54 for women.

Below 0.02: 6 male and 3 female graduates against 359 and 577 non-graduates. Above 0.54: 200 and 206 graduates against 82 and 86.

The low tail rests on a handful of people; they report wide intervals there and rerun everything without the bottom 20%.

(Cheng et al. 2021, Fig. 2, CC BY-NC)

Cheng et al.: the Results (their Figure 3)



A growth curve for log wage on the matched sample, the college effect a cubic spline in $\hat{e}$.

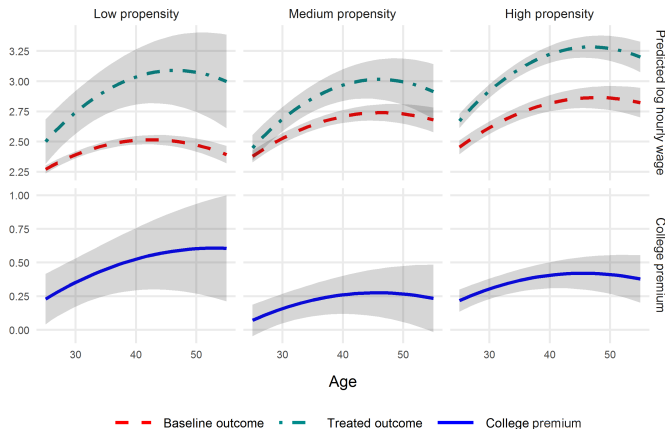
Men: a U. About 0.6, 0.2, and 0.8 at low, median, and high propensity by age 40, deepening with age.

Women: an L. 0.4 at the low end, then 0.2–0.3, flat.

Heterogeneity test: men $\chi^2_6 = 18.66$ ($p = .005$), women 2.34 ($p = .89$).

(Cheng et al. 2021, Fig. 3, CC BY-NC)

Cheng et al.: Where the U Comes From (their Figure 4)



Men at the 20th, 50th, and 80th percentile of $\hat{e}$: predicted wages with and without a degree (top) and their gap (bottom).

Low-propensity graduates escape a flat non-college trajectory; high-propensity graduates ride a steeper one; in the middle the two curves nearly coincide.

For women (their Figure 5) the curves converge with age instead.

(Cheng et al. 2021, Fig. 4, CC BY-NC)

Weighting

Why Weighting, After Matching ★

It is the same operation. Matching's ATT gave each untreated unit in cell x the weight n_{1x}/n_{0x} . Weighting gives it $e(x)/(1 - e(x))$: the same ratio with a model in place of the count. Both average $\tau(x)$ with the target's weights.

What the model buys. A one-armed or empty cell is not dropped: a logit says how $\Pr(D = 1 \mid X)$ moves with X and lends that to every cell. Every unit is used, continuous X and many covariates are no obstacle, one formula serves every target, and it plugs into the doubly robust estimator.

What it costs. Thin overlap does not disappear; it moves. Matching drops a treated unit with no comparison and says so. Weighting keeps it at weight $1/\hat{e} = 100$ and says so only through the variance and the largest weights; and the weights are right only where the model is, including where the data are too thin to check.

The trade: matching narrows the target and says so; weighting keeps the target and pays in variance and model dependence. Trimming $\hat{e}$ to $[0.1, 0.9]$ first makes the two agree, and is common practice.

Recovering the ATT ★

In cell x , n_{1x}/n_{0x} estimates $\Pr(D_i = 1 \mid x)/\Pr(D_i = 0 \mid x) = e(x)/(1 - e(x))$, the *odds* of treatment. So matching's ATT weights were, in population terms:

$$\text{treated: } 1, \quad \text{untreated: } \frac{e(X_i)}{1 - e(X_i)}.$$

Check that the weighted untreated mean is the treated group's counterfactual:

$$\begin{aligned} E\left[\frac{e(X_i)}{1-e(X_i)}(1-D_i)Y_i\right] &= E\left[\frac{e(X_i)}{1-e(X_i)}E[(1-D_i)Y_i(0) \mid X_i]\right] && \text{iterate on } X_i \\ &= E\left[\frac{e(X_i)}{1-e(X_i)}(1-e(X_i))E[Y_i(0) \mid X_i]\right] && \text{ignorability} \\ &= E[e(X_i)E[Y_i(0) \mid X_i]] && \text{cancel} \\ &= E[D_i Y_i(0)] = \pi E[Y_i(0) \mid D_i = 1] && e(X_i) = E[D_i \mid X_i], \text{ iterate} \end{aligned}$$

Divide by $\pi = E[D_i]$, which is what the weights sum to. The treated need no weights: $E[D_i Y_i]/\pi = E[Y_i(1) \mid D_i = 1]$. Subtract: the ATT.

Recovering the ATE ★

Now the target is everyone. Cell x holds n_x units, of which n_{1x} are treated; for the treated to stand in for the whole cell, each must count n_x/n_{1x} times, which estimates $1/e(x)$. Likewise each untreated unit counts $1/(1 - e(x))$:

$$\widehat{\text{ATE}}_{\text{IPW}} = \frac{1}{n} \sum_{i=1}^n \left[\frac{D_i Y_i}{\hat{e}(X_i)} - \frac{(1 - D_i) Y_i}{1 - \hat{e}(X_i)} \right]$$

Rare treated units in a cell count for more, because they stand in for everyone like them who was not treated. The same check as before, treated arm:

$$E \left[\frac{D_i Y_i}{e(X_i)} \right] = E \left[\frac{E[D_i | X_i] E[Y_i(1) | X_i]}{e(X_i)} \right] = E[E[Y_i(1) | X_i]] = E[Y_i(1)],$$

by ignorability given X_i and then iterated expectations; the same for the untreated arm with $1 - e$. ■ This is the Horvitz–Thompson estimator of survey sampling, with treatment assignment as the sampling mechanism.

IPW Is a Weighted Regression

Define the **Horvitz–Thompson transformation**

$$H_i = \frac{D_i}{e(X_i)} - \frac{1 - D_i}{1 - e(X_i)}.$$

Then $E[H_i Y_i] = E[Y_i(1) - Y_i(0)]$: the ATE weights of the last frame in one term. Equivalently:

The coefficient on D_i from a **weighted least squares** regression of Y_i on D_i (with an intercept), weights $1/\hat{e}(X_i)$ for the treated and $1/(1 - \hat{e}(X_i))$ for the controls, is a difference in weighted means with the weights normalized to sum to one in each arm (the Hájek form), rather than the raw average that divides by n .

- Practically convenient: `lm(y ~ d, weights = w)` runs IPW, and covariates can be added for extra protection.
- Conceptually important: weighting and regression are one continuum, and this is the bridge to the doubly robust estimator.

Recovering the ATU, and the Whole Table ★

The ATU mirrors the ATT: the untreated are the target, so they keep weight 1, and the treated are reweighted to look like them with the inverse odds $(1 - e(X_i))/e(X_i)$. Side by side, every weighting is a choice of whom the other arm must resemble:

Target	Weight on treated	Weight on untreated	Who is reweighted
ATE	$1/e(X)$	$1/(1 - e(X))$	both, to the population
ATT	1	$e(X)/(1 - e(X))$	the untreated, to the treated
ATU	$(1 - e(X))/e(X)$	1	the treated, to the untreated
Overlap (ATO)	$1 - e(X)$	$e(X)$	both, to where they overlap

- Each row is a covariate distribution the arms are pushed toward; the ATT row is $\Pr(X \mid D = 1)$, matching's weights from slide 5.
- The last row is new: a target defined by the data rather than by a population. Next frame.

The Overlap Weights, Cell by Cell ★

Cell x holds $n_{1x} = e(x) n_x$ treated and $n_{0x} = (1 - e(x)) n_x$ untreated units. Give each treated unit the weight $1 - e(x) = n_{0x}/n_x$, the share of its cell it can be compared with, and each untreated unit $e(x) = n_{1x}/n_x$. Then in every cell

$$\underbrace{n_{1x} \frac{n_{0x}}{n_x}}_{\text{treated total}} = \frac{n_{1x} n_{0x}}{n_x} = \underbrace{n_{0x} \frac{n_{1x}}{n_x}}_{\text{untreated total}}.$$

- **Balanced by construction:** both arms carry the same total weight in every cell. **Bounded:** no weight exceeds 1.
- **Whom it describes.** Cell x counts in proportion to $n_{1x} n_{0x} / n_x = n_x e(x)(1 - e(x))$, the comparisons it affords: a one-armed cell drops out on its own, a half-and-half cell counts most. The target is “people for whom the data contain a comparison”, the *overlap population*, ATO.
- $n_x e(x)(1 - e(x))$ is slide 6’s regression weighting: ATO is what saturated OLS targeted all along, now chosen deliberately (Li, Morgan, and Zaslavsky 2018).

Why It Works, and When It Does Not ★

- **The mechanism.** Within cell x , ignorability makes the treated a random sample of the cell. Weighting corrects only the *sizes* of those samples so that each arm has the target's cell distribution. Then a weighted mean of Y in an arm is a mean of $Y(d)$ over the target population. That is the whole proof, twice over on the last two frames.
- **It uses every unit.** No pairing, no discarding, and one formula for every estimand; only $e(x)$ has to be estimated.
- **It needs $e(x)$ right.** Matching on cells needed nothing modelled; weighting needs $\Pr(D = 1 \mid X)$, usually a logit. A wrong model gives wrong weights everywhere at once.
- **It needs positivity, loudly.** A treated unit with $\hat{e} = 0.01$ carries the weight of a hundred; the variance goes with it. Where matching dropped a cell, weighting inflates it. Diagnose with the largest weights and the effective sample size, $(\sum_i w_i)^2 / \sum_i w_i^2$; then trim, or change the estimand.

In the Literature: Wodtke, Harding, and Elwert (2011)

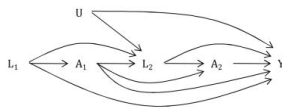
The question. Does growing up in a disadvantaged neighbourhood lower the chance of finishing high school? Earlier studies measured the neighbourhood once, so a recent arrival counted like a lifelong resident, and controlled for family income, marital status, and employment, which the neighbourhood itself changes. Both understate a lifetime of exposure.

The data. PSID children aged 1 in 1968–1978 ($n = 4,154$); 2,093 followed every year to 17 and seen at 20 (834 Black, 1,259 non-Black). Neighbourhood = the tract's disadvantage score (poverty, unemployment, welfare, female-headed households, education, professional jobs) in national quintiles: the treatment is a sequence $A_2, \dots, A_{17}$, each in $\{1, \dots, 5\}$. Outcome: graduation by 20.

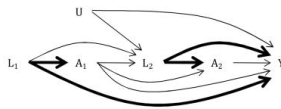
The estimand. 5^{16} possible histories, so a *marginal structural model* on the potential outcomes: $\text{logit } \Pr(Y_{\bar{a}} = 1) = \theta_0 + \theta_1 \frac{1}{16} \sum_k a_k$. θ_1 is the effect of one quintile more disadvantage, on average, over childhood.

Wodtke et al.: Why Regression Fails (their Figure 1)

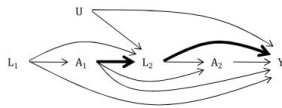
A. Hypothesized causal relationships



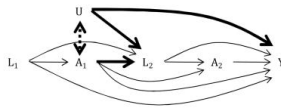
B. Do not condition on $L_k \rightarrow$ confounding



C. Condition on $L_k \rightarrow$ over-control of intermediate pathways



D. Condition on $L_k \rightarrow$ collider-stratification bias



A_1, A_2 : neighbourhood at two waves. L_2 : family income, marital and employment status at wave 2. U : unobserved. Y : graduation. **B.** L_2 confounds A_2 : it must be adjusted for. **C.** L_2 lies on $A_1 \rightarrow L_2 \rightarrow Y$: adjusting removes the early effect that runs through the family (post-treatment control, Week 5). **D.** L_2 is a collider of U and A_1 : adjusting opens $A_1 \leftrightarrow U \rightarrow Y$, though nothing unobserved affects A . So regression is biased whether it adjusts for L_2 or not: the confounder is also a mediator and a collider. (their Fig. 1)

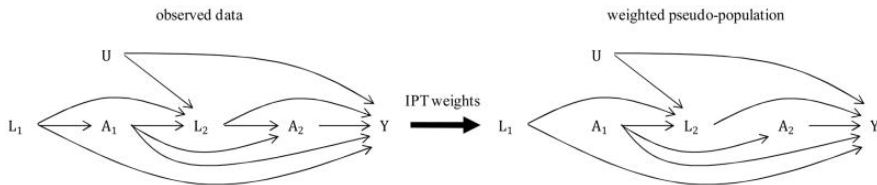
Wodtke et al.: the Weights

Weight each child by the inverse probability of the neighbourhood sequence she actually had, one wave at a time, stabilized:

$$sw_i = \prod_{k=1}^{16} \frac{\Pr(A_k = a_{ki} \mid \bar{A}_{k-1} = \bar{a}_{(k-1)i}, L_0)}{\Pr(A_k = a_{ki} \mid \bar{A}_{k-1} = \bar{a}_{(k-1)i}, \bar{L}_k = \bar{l}_{ki})}$$

- **Denominator:** the probability of this year's quintile given last year's, the baseline covariates, and the family's history up to now. An ordinal logit per race of A_k on A_{k-1} , baseline covariates, the family variables at k and $k-1$, and divorce and job-loss interactions.
- **Numerator:** the same model without the family variables. It cancels nothing that matters and keeps the weights near one (their stabilization).
- **Sixteen factors.** Our one-shot weight $1/e(x)$ becomes a product over years: the same idea, applied every year given the past.

Wodtke et al.: the Pseudo-Population (their Figure 2)



- In the weighted pseudo-population the arrows from L_k into A_k are gone: exposure behaves as if re-randomized each year given the past.
- So the outcome model is an *unadjusted* logit of Y on average exposure, plus baseline covariates only. No L_k in it, so nothing is controlled away and no collider is opened.
- **Assumptions:** no unmeasured confounder of A_k given the history, positivity at every wave, a correct treatment model. Regression needs all three *and* that L be unaffected by past A , which the DAG denies.

Wodtke et al.: Checking the Weights, and the Results

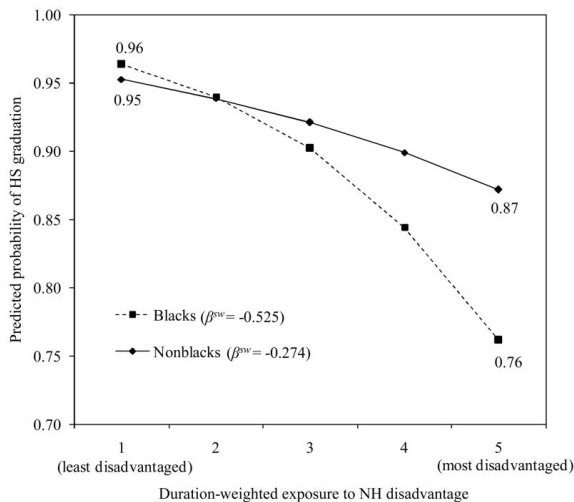
Their Table 4. Stabilized weights: Black children mean 1.03, SD 0.58, 1st and 99th percentiles 0.27 and 4.62; non-Black 1.00, 0.31, 0.32 and 2.54. Centred at one, no monsters; truncated at the 1st and 99th percentiles anyway.

Their Table 5. Log odds of graduation per quintile of average exposure:

	Black ($n = 834$)	non-Black ($n = 1,259$)
unadjusted	-.703 (.170)	-.581 (.109)
regression, family variables as controls	-.416 (.196)	-.212 (.125)
IPT-weighted	-.525 (.190)	-.274 (.128)

- Regression with the family variables as controls halves the effect: it controls away the part that runs through the family. Weighting keeps it. The difference between the rows is the mediated share.
- Fourteen alternative treatment models (more interactions, squares, fewer covariates): $-.507$ to $-.588$ and $-.268$ to $-.294$.

Wodtke et al.: the Results (their Figure 3)



Predicted probability of graduation by average exposure, from the weighted logit with baseline covariates at their race-specific means.

Least to most disadvantaged quintile throughout childhood takes graduation from 96% to 76% for Black children and from 95% to 87% for non-Black children.

About 65% of Black children in the sample, and 9% of non-Black children, spent most of childhood in the most disadvantaged quintile. The exposure itself is unequal; the effect of it, once weighted, is large for both.

(Wodtke, Harding, and Elwert 2011, Fig. 3)

Doubly Robust

G-Computation: the Other Half ★

Instead of modelling treatment, model the outcome. Under ignorability and common support, fit $m_d(x) = E[Y_i \mid D_i = d, X_i = x]$ for $d = 0, 1$, then *impute both potential outcomes for everyone* and average:

$$\hat{\tau}_{\text{g-comp}} = \frac{1}{n} \sum_{i=1}^n [\hat{m}_1(X_i) - \hat{m}_0(X_i)].$$

- Week 4's predicted-probability procedure. One common-slope linear model returns τ_R ; separate models per arm average $\tau(x)$ with the ATE weights $\Pr(X_i = x)$, and slide 6's variance weighting is gone.
- Uses the whole sample; efficient when the outcome model is right; **extrapolates silently** when it is not, since $\hat{m}_1(x)$ is computed even where nobody was treated (the anatomy shows how badly).
- The symmetry: IPW models $\Pr(D \mid X)$ and never touches Y ; g-computation models $E[Y \mid D, X]$ and never touches assignment.

Why Not Both? ★

IPW is consistent only if $e(X_i)$ is right. G-computation is consistent only if $m_d(X_i)$ is right. Combine them so that *either* suffices.

Given the two nuisance functions $m_d(x) = E[Y_i \mid D_i = d, X_i = x]$ and $e(x) = \Pr(D_i = 1 \mid X_i = x)$, the **augmented IPW** estimand is

$$\tau_{\text{AIPW}} = E \left[\underbrace{m_1(X_i) - m_0(X_i)}_{\text{g-computation}} + \underbrace{\frac{D_i(Y_i - m_1(X_i))}{e(X_i)} - \frac{(1 - D_i)(Y_i - m_0(X_i))}{1 - e(X_i)}}_{\text{IPW-weighted residuals}} \right]$$

Double robustness

$\tau_{\text{AIPW}} = E[Y_i(1) - Y_i(0)]$ if **either** m_d **or** e is correctly specified, not necessarily both. And when both are right, AIPW attains the semiparametric efficiency bound.

Case 1: the Outcome Model Is Correct ★

Suppose m_1 is right but e is arbitrary. Look at the augmentation term for the treated arm, conditioning on X_i :

$$\begin{aligned} E\left[\frac{D_i}{e(X_i)}(Y_i - m_1(X_i)) \mid X_i\right] &= \frac{1}{e(X_i)} \Pr(D_i = 1 \mid X_i) E[Y_i - m_1(X_i) \mid D_i = 1, X_i] \\ &= \frac{1}{e(X_i)} E[D_i \mid X_i] \cdot \underbrace{0}_{\text{by definition of } m_1} = 0. \end{aligned}$$

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The same argument gives $E\left[\frac{1-D_i}{1-e(X_i)}(Y_i - m_0(X_i)) \mid X_i\right] = 0$. Hence

$$\tau_{\text{AIPW}} = E[m_1(X_i) - m_0(X_i)] = E[Y_i(1) - Y_i(0)].$$

The whole augmentation term vanishes, whatever e is. Note that this did not use a correct propensity score anywhere.

Case 2: the Propensity Score Is Correct ★

Now suppose e is right but m_1 is some wrong function. Then

$$E\left[\frac{D_i}{e(X_i)}(Y_i - m_1(X_i)) \mid X_i\right] = \frac{1}{e(X_i)}E[D_i Y_i \mid X_i] - m_1(X_i).$$

Since $E[D_i Y_i \mid X_i] = e(X_i) E[Y_i(1) \mid X_i]$ by ignorability, this is

$$E[Y_i(1) \mid X_i] - m_1(X_i).$$

Case 2: the Propensity Score Is Correct ★

Now suppose e is right but m_1 is some wrong function. Then

$$E\left[\frac{D_i}{e(X_i)}(Y_i - m_1(X_i)) \mid X_i\right] = \frac{1}{e(X_i)}E[D_i Y_i \mid X_i] - m_1(X_i).$$

Since $E[D_i Y_i \mid X_i] = e(X_i) E[Y_i(1) \mid X_i]$ by ignorability, this is

$$E[Y_i(1) \mid X_i] - m_1(X_i).$$

Symmetrically the control term contributes $-(E[Y_i(0) \mid X_i] - m_0(X_i))$. Adding everything up, the wrong m_1, m_0 cancel exactly:

$$\tau_{\text{AIPW}} = E\left[E[Y_i(1) \mid X_i] - E[Y_i(0) \mid X_i]\right] = E[Y_i(1) - Y_i(0)]. \quad \blacksquare$$

Two independent routes to the same answer. That is the insurance.

Why the Cancellation Matters Beyond This Week

Look again at what happened in the two cases. A first-order error in *one* nuisance function was wiped out by the other term. That is not a coincidence of AIPW; it is a property the estimating equation was built to have.

The consequence

The bias of AIPW is roughly the *product* of the two nuisance errors, not their sum. So if both $\hat{e}$ and $\hat{m}_d$ are merely *approximately* right, the bias is second-order small, much smaller than either model's own error.

- This is why doubly robust estimators are the default in modern practice even when nobody believes either model exactly.
- It is also what eventually licenses estimating $\hat{e}$ and $\hat{m}_d$ with flexible, machine-learning methods rather than logit and OLS. **That is a Week 13 topic**; the property has a name (*Neyman orthogonality*) and we will not need it before then.

In the Literature: Lundberg (2024)

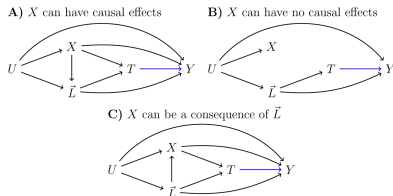
The question. People whose fathers held professional jobs earn more. How much of that gap would remain if we *intervened* on the respondent's own occupation? The category (father's class) is never the treatment; the treatment is something we could assign. That is the *gap-closing estimand*: the disparity across categories that would persist under a counterfactual assignment of the treatment to a sample from the population.

The data. General Social Survey 1975–2018, $n = 12,328$. Category X : father in a professional occupation (EGP class I), 10.2%. Treatment T : respondent in a professional occupation, 9.6%. Outcome: log annual income in 1986 dollars. Covariates L : race, sex, age, highest degree. Survey weights throughout.

The descriptive gap is 0.32 log points. A natural story: 24% of professional-origin respondents reach professional jobs themselves, against 8% of working-class-origin respondents. Would equalizing destinations close the gap?

Lundberg: Identification (their Figure 3)

A gap-closing estimand is identified under a wide range of assumptions about the social category X (e.g. race, class, gender).



Above, there is no causal effect of X (B), or it is not identified (A, C) due to the backdoor path $X \leftarrow U \rightarrow Y$ through unobserved U . The gap-closing estimand is nonetheless identified.

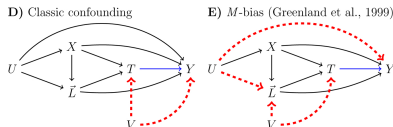
The blue arrow $T \rightarrow Y$ is the only effect that must be identified.

A, B, C. Whether X causes things, causes nothing, or is caused by the covariates makes no difference: every back-door path into T runs through X or L , both observed, so conditioning on them identifies $T \rightarrow Y$ (Week 5). The effect of X itself is not identified and never needed.

D, E. An unobserved V into T and Y breaks it, as ordinary confounding or through the M structure. This is the week's conditional ignorability, stated for T within levels of X and L .

(Lundberg 2024, Fig. 3)

A gap-closing estimand is **not** identified when $T \rightarrow Y$ is not identified.



Lundberg: the Estimator (their Equation 8)

$$\hat{\theta}_x^{\text{Outcome}}(\pi) = \frac{1}{\eta_x} \sum_{i: X_i=x} \eta_i \sum_t \hat{\pi}(t, \vec{L}_i) \hat{g}(t, X_i, \vec{L}_i)$$

Sample weight
Counterfactual probability of treatment t
Predicted outcome under treatment t

↘
↓
↙

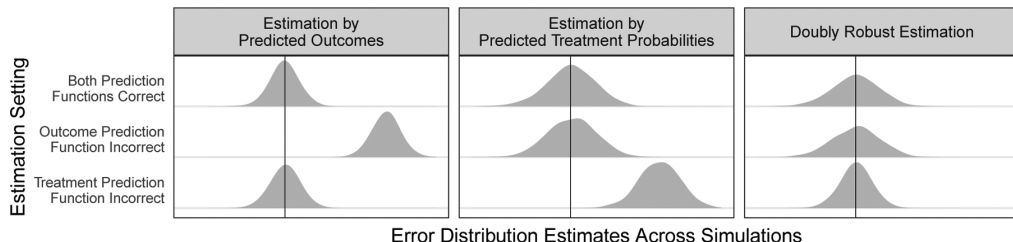
↑
↗
↖
↗↖

Estimate from predicted outcomes
Sum over all sampled units i in category x
Sum over all treatment values t
Category and covariates as observed

Exactly our AIPW, one category x at a time:

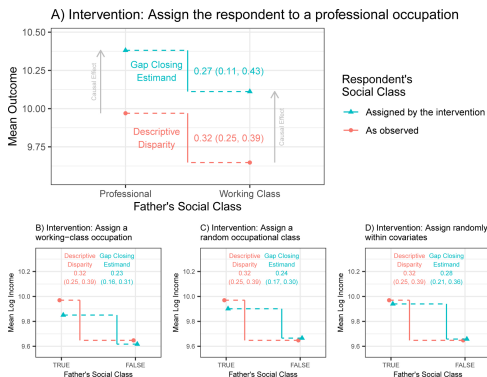
- $\hat{g}$: the outcome model, OLS of log income on T , X , L . $\hat{m}$: the treatment model, a logit. $\hat{w}_i$: the inverse probability of the treatment received, times the survey weight.
- Start from g-computation (predict everyone's income under $T = 1$, average within the category), then subtract the weighted average residual among units observed under $T = 1$: what the outcome model gets wrong, reweighted to the target.
- Consistent if either $\hat{g}$ or $\hat{m}$ is right: our Cases 1 and 2.

Lundberg: Does It Work? (their Figure 4)



His simulation, 1,000 draws of $n = 500$, the truth marked by the vertical line. Row 2: the outcome model lacks an interaction, so g-computation (left) is off. Row 3: the treatment model lacks a quadratic, so weighting (middle) is off. The doubly robust estimate (right) stays centred on the truth in every row. It is wider than g-computation when both models are right: the price of insurance.

Lundberg: the Results (their Figure 6)



Panel A. Assign everyone a professional occupation: both groups' mean log income rises by about the same amount, so the gap moves only from 0.32 (0.25, 0.39) to 0.27 (0.11, 0.43), 84% of its size.

B, C, D. A working-class job: 0.23. A random class in population proportions: 0.24. A random class within covariate groups: 0.28.

No intervention on one's own class closes the gap by one's father's class. The status-attainment story, in which one's own achievement overwrites origins, does not survive.

Doubly robust with parametric $\hat{g}$ and $\hat{m}$; intervals by balanced repeated replication; machine-learning versions in his appendix agree.

An Honest Warning ★

Double robustness is often oversold.

- “Either model correct” is a low bar in theory and an unreachable one in practice: both models are almost certainly wrong, and then AIPW is inconsistent too.
- What it really buys is **second-order bias**: the bias is the *product* of the two models’ errors, so it is small when both are approximately right. That is genuinely valuable, and it is the honest version of the claim.
- With extreme weights, AIPW can be more unstable than either component alone (Kang and Schafer 2007).
- **None of this addresses unobserved confounding**. Everything today assumes conditional ignorability, and no diagnostic can test it.

Sensitivity Analysis: Taking the Assumption Seriously ★

Since ignorability is untestable, ask instead: *how strong would an unobserved confounder have to be to overturn my conclusion?*

- **Rosenbaum bounds:** how much would two matched units have to differ in their odds of treatment before the inference flips? (DiPrete and Gangl 2004: unemployment insurance and re-employment)
- **Partial R^2 / robustness value** (Cinelli and Hazlett 2020): express the required confounder strength as the share of residual variation in D and in Y it would have to explain, then benchmark it against an *observed* covariate, “the confounder would need to be three times as strong as parental education.”
- **E -value:** the minimum risk ratio a confounder would need with both treatment and outcome.

Report one. It converts an untestable assumption into a quantitative claim your reader can argue with.

Choosing Among the Methods

Method	Models	Main weakness
Exact / CEM matching	nothing	discards data; infeasible with many covariates
NN / Mahalanobis matching	a distance	bias with continuous X ; variance from reuse
PS matching	$\Pr(D X)$	sensitive to the PS model; discards controls
IPW	$\Pr(D X)$	extreme weights when overlap is poor
Regression / g-computation	$E[Y D, X]$	silent extrapolation; own weighting
AIPW	both	unstable weights; still needs ignorability

- If the answers differ a lot, that is worth chasing down. As slide 6 said, systematic differences can mean the cell-level effects $\tau(x)$ are heterogeneous and the methods weight them differently, but they can equally reflect a misspecified model, poor overlap, or sampling noise. Distinguish these before concluding anything.

Choosing Among the Methods (cont.)

- Report the primary estimator you pre-committed to, plus the others as robustness, never the reverse.

A Workflow ★

1. State the estimand (ATE? ATT? ATO? on which population?) before touching data.
2. Draw the DAG. Decide what to condition on, and what *not* to.
3. Estimate the propensity score. Look at overlap. Trim or redefine if needed and say so.
4. Match or weight. Check balance with standardized mean differences and density plots, **never** with t -tests.
5. Estimate with a doubly robust estimator; report matching, weighting, and regression alongside.
6. Get standard errors that respect the design: bootstrap the whole estimation pipeline for weighting and regression estimators, and cluster where treatment was assigned. **Do not bootstrap nearest-neighbour matching**, the ordinary bootstrap is invalid for it (Abadie and Imbens 2008); use their analytic variance instead.
7. Report a sensitivity analysis.

Anatomy

What OLS Does With the Cell Effects: Goal and Plan

Slide 6 asserted that OLS on D_i and a saturated X_i returns a variance-weighted average of the $\tau(x)$. Now the derivation. From slide 5,

$$\begin{aligned}\tau(x) &= E[Y_i \mid D_i = 1, X_i = x] - E[Y_i \mid D_i = 0, X_i = x] \\ \text{ATE} &= \sum_x \tau(x) \Pr(X_i = x), \quad \text{ATT} = \sum_x \tau(x) \Pr(X_i = x \mid D_i = 1).\end{aligned}$$

1. Rewrite the ATT weights in terms of $e(x)$: one line of Bayes' rule.
2. Write the OLS coefficient by Frisch–Waugh–Lovell.
3. Simplify its numerator.
4. Read off the weights, and compare them with matching's.
5. Two consequences, the continuous case, and Angrist (1998) as the empirical check.

Anatomy of the Matching Estimand

Apply Bayes' rule to the ATT weights:

$$\Pr(X_i = x \mid D_i = 1) = \frac{\Pr(D_i = 1 \mid X_i = x) \Pr(X_i = x)}{\Pr(D_i = 1)}.$$

Anatomy of the Matching Estimand

Apply Bayes' rule to the ATT weights:

$$\Pr(X_i = x \mid D_i = 1) = \frac{\Pr(D_i = 1 \mid X_i = x) \Pr(X_i = x)}{\Pr(D_i = 1)}.$$

Substituting,

$$\text{ATT} = \frac{\sum_x \tau(x) \Pr(D_i = 1 \mid X_i = x) \Pr(X_i = x)}{\sum_x \Pr(D_i = 1 \mid X_i = x) \Pr(X_i = x)}$$

The matching-based ATT weights each cell effect in proportion to the probability of treatment in that cell; that is, in proportion to the propensity score $e(x)$. Cells where treatment is common count for more, which is exactly what “the effect on the treated” should mean.

Anatomy of Regression Adjustment: Setup ★

Now do the same accounting for regression. To avoid conflating the weighting question with a functional-form question, specify a model *saturated in* X_i :

$$Y_i = \sum_x \mathbb{1}(X_i = x) \alpha_x + \tau_R D_i + \epsilon_i.$$

By Frisch–Waugh–Lovell (Week 1), the coefficient on D_i equals the slope of Y_i on the part of D_i the cell dummies cannot predict. Because the model is saturated, that projection is the cell mean of D_i , $e(X_i)$, so

$$\check{D}_i \equiv D_i - e(X_i) = \begin{cases} 1 - e(X_i) & \text{treated} \\ -e(X_i) & \text{untreated} \end{cases} \quad \text{and} \quad \tau_R = \frac{\text{Cov}(Y_i, \check{D}_i)}{\text{Var}(\check{D}_i)}.$$

$\check{D}_i$ is the *surprise* in treatment given the cell: large for a treated unit in a cell where treatment is rare. Next: rewrite the covariance so that the cell effects $\tau(x)$ appear.

Anatomy of Regression Adjustment: the Covariance ★

Two facts about $\check{D}_i$ do all the work: its mean is zero, and it is a function of (D_i, X_i) only.

$$E[\check{D}_i] = E[D_i] - E[E[D_i | X_i]] = 0$$

iterated expectations

$$\text{Cov}(Y_i, \check{D}_i) = E[\check{D}_i Y_i] - E[\check{D}_i] E[Y_i] = E[\check{D}_i Y_i]$$

definition; second term is 0

$$= E[E[\check{D}_i Y_i | D_i, X_i]]$$

iterate on (D_i, X_i)

$$= E[\check{D}_i E[Y_i | D_i, X_i]]$$

$\check{D}_i$ is fixed once (D_i, X_i) are

$$\text{Var}(\check{D}_i) = E[\check{D}_i^2]$$

mean zero

The last step is the one to see: inside $E[\cdot | D_i, X_i]$, anything built from D_i and X_i is a constant and comes out. So

$$\tau_R = \frac{E[\check{D}_i E[Y_i | D_i, X_i]]}{E[\check{D}_i^2]},$$

and the numerator now contains $E[Y_i | D_i, X_i]$, which is where $\tau(x)$ lives. Next frame.

Anatomy of Regression Adjustment: the Numerator ★

Because D_i is binary, the conditional expectation decomposes exactly:

$$\begin{aligned} E[Y_i \mid D_i, X_i] &= E[Y_i \mid D_i = 0, X_i] + D_i \left(E[Y_i \mid D_i = 1, X_i] - E[Y_i \mid D_i = 0, X_i] \right) \\ &= E[Y_i \mid D_i = 0, X_i] + D_i \tau(X_i). \end{aligned}$$

Anatomy of Regression Adjustment: the Numerator ★

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Substitute into the numerator:

$$\begin{aligned} &E \left[(D_i - E[D_i \mid X_i]) (E[Y_i \mid D_i = 0, X_i] + D_i \tau(X_i)) \right] \\ &= \underbrace{E \left[(D_i - E[D_i \mid X_i]) E[Y_i \mid D_i = 0, X_i] \right]}_{=0 \text{ (a function of } X_i \text{ only)}} + E \left[(D_i - E[D_i \mid X_i]) D_i \tau(X_i) \right] \\ &= E \left[(D_i - E[D_i \mid X_i])^2 \tau(X_i) \right], \end{aligned}$$

where the last step uses $E[(D_i - E[D_i \mid X_i]) D_i \mid X_i] = \text{Var}(D_i \mid X_i)$.

Anatomy of Regression Adjustment: the Result

Putting numerator and denominator together and applying iterated expectations:

$$\tau_R = \frac{E[(D_i - E[D_i | X_i])^2 \tau(X_i)]}{E[(D_i - E[D_i | X_i])^2]} = \boxed{\frac{E[\sigma_D^2(X_i) \tau(X_i)]}{E[\sigma_D^2(X_i)]}}$$

where $\sigma_D^2(x) \equiv \text{Var}(D_i | X_i = x)$. Since D_i is binary, $\sigma_D^2(x) = e(x)(1 - e(x))$, so

$$\tau_R = \frac{\sum_x \tau(x) e(x)(1 - e(x)) \Pr(X_i = x)}{\sum_x e(x)(1 - e(x)) \Pr(X_i = x)}.$$

Regression weights each cell by the *conditional variance of treatment*. Compare with matching, which weighted by $e(x)$ itself. Cells with $e(x) \approx 0.5$ dominate the regression, and cells that are nearly all treated or nearly all control contribute almost nothing to τ_R . Matching behaves differently at the top end, which is the next slide.

Two Consequences Worth Stating Plainly

- **The two weightings answer different questions** even under identical assumptions and with the same saturated specification. Matching is a *method*, not an estimand, and can be pointed at the ATE or the ATT by changing the cell weights; the ATT weighting above is the one that necessarily differs from τ_R . Neither τ_R nor the ATT is “the” ATE unless effects are homogeneous.
- **The regression weight $e(x)(1 - e(x))$ vanishes at $e(x) \in \{0, 1\}$** , so cells with no comparison units drop out of τ_R silently. **Positivity failures do not produce an error message; they produce a quietly redefined estimand.**
- Matching is different: a cell with $e(x) \approx 1$ carries a *large* ATT weight. But at $e(x) = 1$ there is no control unit, so $\tau(x)$ is simply not identified; the estimator has nothing to average, and the honest response is to say which cells you dropped.

Two Consequences Worth Stating Plainly (cont.)

With continuous covariates

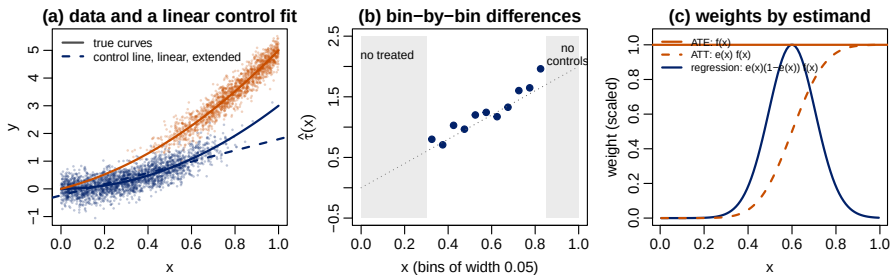
The anatomy still holds, as an integral:

$$\tau_R = \frac{\int \tau(x) e(x) (1 - e(x)) f(x) dx}{\int e(x) (1 - e(x)) f(x) dx}.$$

Continuous X : the Cells Become Bins

With continuous X no two units share a value: “the cell x ” becomes a bin $(x - h, x + h)$, and positivity becomes

$E[Y_i(d) \mid X_i = x] = \lim_{h \rightarrow 0} E[Y_i(d) \mid X_i \in (x - h, x + h)]$, both arms *near* every x . A simulation (figs/w6_fuzzy.R):



x uniform, $e(x)$ rising steeply around 0.6, control surface $3x^2$, effect $\tau(x) = 2x$: ATE 1.00, ATT 1.57. Bins of width 0.05 with at least five units per arm: 11 of 20, $x \in [0.30, 0.85]$, 54% of the sample, 629 of 1,611 treated unmatched.

Matching and Regression on the Overlap

	ATE	ATT
truth, whole sample	1.00	1.57
truth, overlap population [0.30, 0.85]	1.15	
binned matching, overlap only	1.24	1.50
OLS coefficient, D and x	1.56	
OLS coefficient, D , x , x^2	1.19	

- **Matching answers about the overlap.** Panel (b): $\hat{\tau}(x)$ exists only where both arms are dense. Averaged, it gives the overlap population's effect (1.24 vs. 1.15: within-bin imbalance, treated sit higher in each bin) and the matchable treated's (1.50). It says nothing about the 629 treated above 0.85.
- **The OLS coefficient does not extrapolate; it reweights.** Panel (c): $e(x)(1 - e(x))$ puts 83% of its weight on $x \in [0.45, 0.75]$, so with the right curvature $\tau_R \approx 2 \times 0.6 = 1.19$: fuzzy matching with weights it chose. With the wrong curvature, 1.56.

G-Computation Extrapolates

Separate outcome models per arm, then $\hat{m}_1(X_i) - \hat{m}_0(X_i)$ for everyone. That needs $\hat{m}_0$ at $x = 0.95$, where exactly one control exists. Panel (a) shows what the fitted line does there:

control model	$\hat{m}_0(0.95)$	ATE	ATT
truth	2.71	1.00	1.57
linear in x	1.69	0.90	2.13
quadratic in x	2.66	1.08	1.60

- The linear line runs under the true curve where the treated live, so every treated unit above 0.85 is credited with too large a gain: ATT 2.13 instead of 1.57. Nothing in the data can flag it; no control is there to disagree.
- The quadratic happens to be right and recovers everything. The gap between the rows is a modelling choice made where the data are silent.
- The deepest reason to look at overlap before running anything.

Does It Matter Empirically? Angrist (1998)

Effect of *voluntary* military service on 1988–91 Social Security earnings, among men who applied to enter the armed forces 1979–82. Matching and regression both control for year of birth, education at application, and AFQT score.

	Mean earnings	Difference in means	Matching estimate	Regression estimate	Regression – matching
Whites	14,537	1,233.4 (60.3)	–197.2 (70.5)	–88.8 (62.5)	108.4 (28.5)
Non-whites	11,664	2,449.1 (47.4)	839.7 (62.7)	1,074.4 (50.7)	234.7 (32.5)

Adapted from Angrist (1998), Tables II and V. $n = 128,968$ whites, 175,262 non-whites.

Does It Matter Empirically? Angrist (1998) (cont.)

- Adjustment reverses the sign for whites, selection into service is strongly positive on earnings potential.
- Matching and regression differ by about 100–235 dollars, and the *difference* is statistically significant. Small, but exactly the magnitude the weighting anatomy predicts.

Heckman

What Today Could Not Touch ★

Slide 4's gap, baseline selection plus selection on the effect, was made zero all day by conditioning on X . If units select on something unmeasured, it is not zero, and no diagnostic today can see it. Heckman (1978, 1979): instead of assuming it away, model *who gets treated* and *how their unobservables relate to Y* , then compute it.

$$D_i = \mathbb{1}[Z_i'\gamma + u_i > 0] \quad \text{selection rule: a probit, Week 4}$$

$$Y_i = X_i'\beta + \tau D_i + \epsilon_i \quad \text{outcome, one constant effect } \tau$$

with (u_i, ϵ_i) bivariate normal, $\text{Var}(u_i) = 1$, $\text{Var}(\epsilon_i) = \sigma^2$, $\text{Cov}(u_i, \epsilon_i) = \rho\sigma$.

- $\rho \neq 0$ is selection on unobservables: the same unmeasured thing pushes a unit into treatment and moves its outcome.
- Z_i contains X_i and, ideally, one variable that shifts selection but not Y_i ; why “ideally” comes in two frames.
- A constant τ drops selection on the effect: Heckman's simplification, not a result.

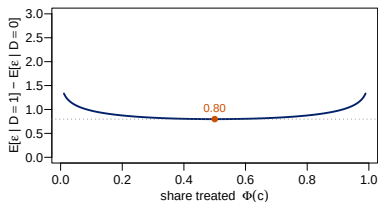
The Gap Has a Formula ★

Bivariate normality gives $E[\epsilon_i | u_i] = \rho\sigma u_i$ (slope Cov / Var). Write $c_i = Z_i'\gamma$, so $D_i = 1$ means $u_i > -c_i$.

$$\begin{aligned} E[u_i | u_i > -c_i] &= \frac{\int_{-c_i}^{\infty} u \phi(u) du}{\Pr(u_i > -c_i)} && \text{conditional mean} \\ &= \frac{\phi(-c_i)}{\Phi(c_i)} = \frac{\phi(c_i)}{\Phi(c_i)} && \phi'(u) = -u \phi(u) \end{aligned}$$

Below $-c_i$ the same integral gives $-\phi(c_i)/[1 - \Phi(c_i)]$.
Multiply both by $\rho\sigma$ and subtract:

$$\begin{aligned} E[\epsilon_i | D_i = 1, Z_i] - E[\epsilon_i | D_i = 0, Z_i] \\ = \rho\sigma \left[\frac{\phi(c_i)}{\Phi(c_i)} + \frac{\phi(c_i)}{1 - \Phi(c_i)} \right] = \rho\sigma \frac{\phi(c_i)}{\Phi(c_i) [1 - \Phi(c_i)]} \end{aligned}$$



The gap at $\rho\sigma = .5$ by treated share: never zero, smallest at one half.

Worked number. $\rho = .5$,
 $\sigma = 1$, $c = 0$ (half treated):
 $.5 \times .399/.25 = 0.80$. With $\tau = 1$
the naive gap should be 1.80;
 10^6 simulated draws give 1.797.

Heckman's Two Steps ★

The omitted variable in the OLS of Y on X and D is a *known function* of c_i times one unknown constant $\rho\sigma$. Estimate the function, then put it into the regression.

1. Probit of D_i on Z_i gives $\hat{\gamma}$ and $\hat{c}_i = Z_i' \hat{\gamma}$.
2. Build the correction term, one expression for both groups:
 $\hat{\lambda}_i = D_i \phi(\hat{c}_i) / \Phi(\hat{c}_i) - (1 - D_i) \phi(\hat{c}_i) / [1 - \Phi(\hat{c}_i)]$, so that $E[\epsilon_i | D_i, Z_i] = \rho\sigma \lambda_i$.
3. OLS of Y_i on $X_i, D_i, \hat{\lambda}_i$. The coefficient on D_i is $\hat{\tau}$; the coefficient on $\hat{\lambda}_i$ estimates $\rho\sigma$, and its test against zero is a test of selection on unobservables.

Simulation, $n = 2,000$, 1,000 replications, true $\tau = 1$, $\rho\sigma = .5$: OLS on X and D averages 1.574; the two-step averages 1.004, with 0.498 on $\hat{\lambda}_i$.

- Step-2 OLS standard errors are wrong: $\hat{\lambda}_i$ is estimated and ϵ_i is heteroskedastic given D_i . Bootstrap both steps, or fit both equations by maximum likelihood (`sampleSelection::treatReg`).
- Wage version (Heckman 1979): Y_i observed only when $D_i = 1$. Same recipe on the $D_i = 1$ rows, where $\hat{\lambda}_i = \phi/\Phi$ is the “inverse Mills ratio”.

What the Correction Costs ★

Estimator of $\tau = 1$	normal errors		skewed errors	
	mean	SD	mean	SD
OLS on X, D	1.574	.043	1.448	.044
Two-step, Z has an excluded variable	1.004	.083	0.958	.072
Two-step, $Z = X$ only	0.954	1.907	-0.166	1.464

$n = 2,000, 1,000$ replications; skewed = centred χ_1^2 shocks, same ρ . `figs/w6_heckman.R`

1. **An exclusion restriction.** With $Z = X$, $\hat{\lambda}_i$ is a smooth function of X_i and D_i : regressing it on them gives $R^2 = .999$ (vs. .667 with Z). Only the curvature of ϕ/Φ identifies τ , and the SD is 23 times larger. The excluded variable must move treatment and have no direct effect on Y : **an instrument**, next week's word.
2. **Bivariate normality.** The gap formula is the normal formula. With an excluded variable, skewed errors cost little here (0.96 for 1); without one, only the normal shape identifies τ , and it returns -0.17 . (Puhani 2000)

Heckman in Sociological Research ★

Mare and Winship (1984, *ASR*). Employment among out-of-school young men, black and white, 1964–1981. That population is itself selected, by school enrollment and military enlistment, differently by race and over time; enrollment and enlistment become the selection equation, and part of the employment trend is composition.

Manski, Sandefur, McLanahan, and Powers (1992, *JASA*). Family structure in adolescence and high school graduation: a nonintact family may hurt, or an unobserved process may produce both. Parametric selection-model estimates are reported next to nonparametric bounds (Week 5), so the reader sees how much of the answer the normality assumption supplies.

Mulligan and Rubinstein (2008, *QJE*). The wage version, in economics but on our question. Selection of women into full-time work turned from negative in the 1970s to positive in the 1990s (exclusion restriction: young children by marital status); holding composition fixed, most of the apparent narrowing of the gender wage gap is a change in who works.

Critiques, and the Bridge to Week 7 ★

- **Berk (1983, *ASR*)** brought it to sociology: who is arrested, employed, or surveyed is a social outcome, so samples are selected on what we study.
- **Winship and Mare (1992, *Annual Review*)** survey its uses in labor markets, schooling, courts, and mobility: results are sensitive to how selection is assumed to occur.
- **Stolzenberg and Relles (1997, *ASR*)** decompose the bias so its likely size can be judged before correcting.
- **Bushway, Johnson, and Slocum (2007)** review criminology's sentencing studies: the two-step run routinely with the same regressors in both equations, the $R^2 = .999$ row.

What survives. The gap is real and has a shape. Fixing it needs a variable that moves treatment and nothing else. Next week keeps that variable, drops normality and the constant τ , and pays with a different estimand.

Looking Ahead

- Everything today but the last section rested on **conditional ignorability**: the claim that we measured every confounder. Matching, weighting, and AIPW make that assumption easier to implement; they do not make it more likely to be true. Heckman trades it for normality and an exclusion restriction.
- The anatomy results are the transferable lesson: *every* adjustment method is a weighting of cell-level effects, and knowing the weights tells you what your number means.
- **Weeks 7–11** abandon ignorability and look for special structure (an instrument, a discontinuity, parallel trends) that identifies an effect *despite* unobserved confounding.
- **Week 13** returns to today's AIPW estimator and asks what changes when the covariate set is large enough that logit and OLS will not do.

Looking Ahead (cont.)

Problem Set 3

Assigned today, due Monday October 19, the Monday *after* the midterm week, so it does not compete with exam preparation. Covers Weeks 5–6.

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